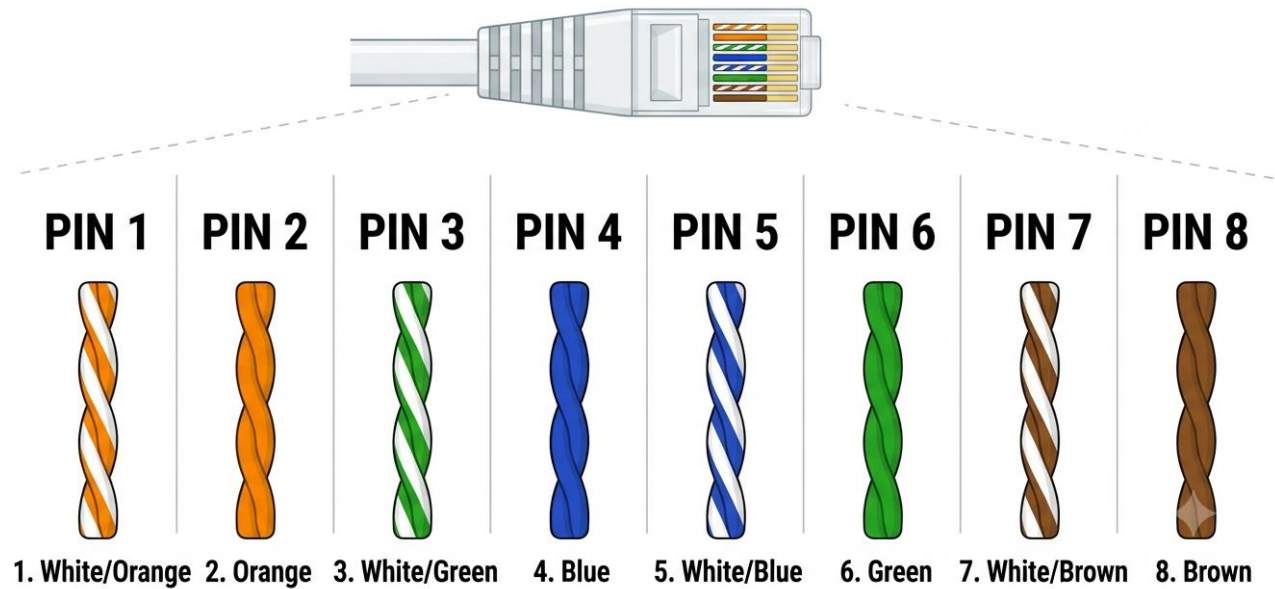


Assignment: Network Cable and Connectors

Task 1: Cat6 Ethernet Cable and T568B Standard

T568B COLOR CODE



Pin	Wire Color
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

Task 2: Types of Network Cables

Cable Type	Real-world Use	Why it is used
Cat5e	Wi-Fi router to laptop	It provides a stable wired network connection at good speed.
Cat6	Zomato POS terminal	It supports higher speeds and reliable connections for payment and order systems.
Fiber Optic	IPL streaming server	It provides very high speed and can carry large amounts of data over long distances.

Task 3: Crimping an RJ-45 Connector

- Cut the network cable to the required length.
- Remove a small part of the outer cable cover.
- Separate and straighten the eight wires.
- Arrange the wires in T568B order.
- Cut the wire ends evenly.
- Insert the wires fully into the RJ-45 connector.
- Check that all wires are in the correct pin order.
- Use a crimping tool to firmly attach the connector.
- Test the cable using a cable tester.

Two common mistakes to avoid:

- Mixing up the wire color order.
- Not pushing all eight wires fully into the RJ-45 connector before crimping.

Task 4: Gaming Zone for an IPL Screening Event

For connecting gaming consoles, screens, and network devices, I would use Cat6 Ethernet cables with RJ-45 connectors.

Cat6 is suitable because it supports high-speed data transfer and gives a more stable connection than Wi-Fi. It can be used to connect gaming PCs or consoles to a network switch or router.

For the main network connection between the event area and a distant server or network room, fiber optic cable can be used because it provides high speed and works well over longer distances.

Suggested setup: Gaming Consoles → Cat6 + RJ-45 → Network Switch → Cat6/Fiber → Router/Server

This setup can provide reliable connections with low network delay during the event.

Task 5: Network Cable or Connector Photo

Photo Placeholder: Insert a photo taken from my home/college showing a Cat6 Ethernet cable with an RJ-45 connector connected to a Wi-Fi router.

Description: The cable shown is a Cat6 Ethernet cable with an RJ-45 connector. It connects the Wi-Fi router to a laptop or another network device. The wired connection provides a stable and reliable network connection.

Image Prompt: Create a realistic photo of a blue Cat6 Ethernet cable with a transparent RJ-45 connector plugged into the LAN port of a home Wi-Fi router. Show the router, Ethernet port, cable, and connector clearly. Natural home/college environment, realistic lighting, suitable for a student networking assignment.